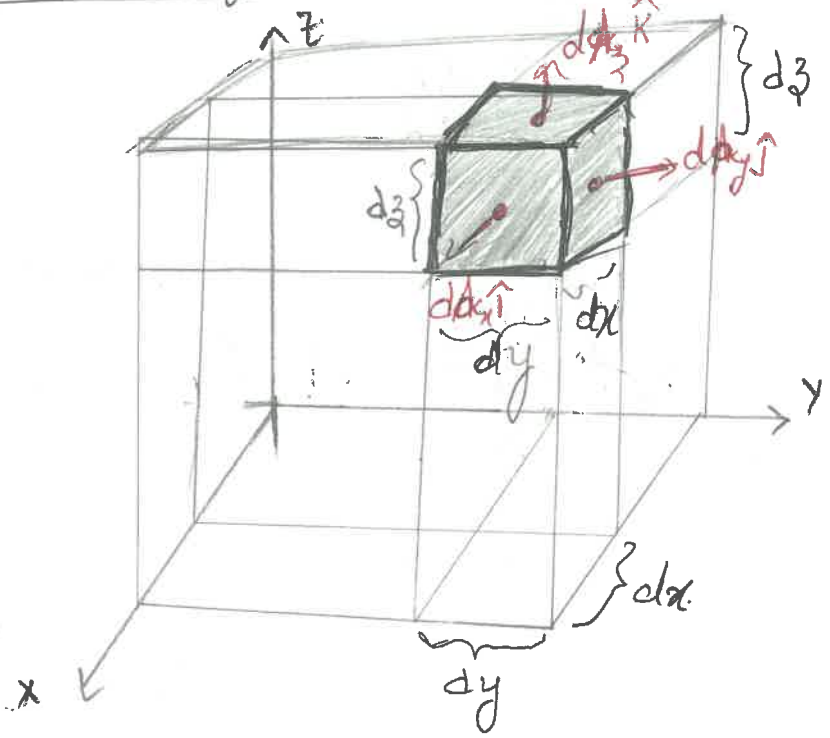
Coordinate Systems: Line, Area & Volume Elements

1)



Cartesian Coordinates.

$$d\vec{l} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

$dA_x = dy \cdot dz\,\hat{i}$ or $-dy\,dz\,\hat{i}$ (for opposite face)

$dA_y = dx\,dz\,\hat{j}$ or $-dx\,dz\,\hat{j}$ (for opposite face)

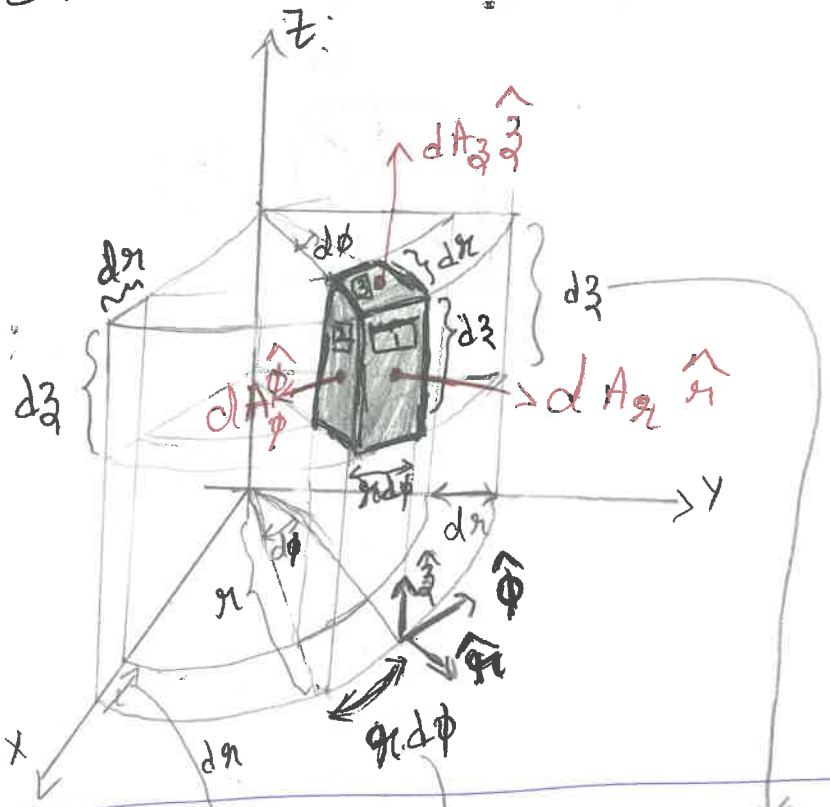
$dA_z = dx\,dy\,\hat{k}$ or $-dx\,dy\,\hat{k}$ (for opposite face)

$$d\tau = d(Vol) = dx\,dy\,dz \quad \rightarrow \text{Volume element}$$

$$d\vec{A} = dy\,dz\,\hat{i} + dx\,dz\,\hat{j} + dx\,dy\,\hat{k} \quad \rightarrow \text{Area element}$$

$$d\vec{l} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k} \quad \rightarrow \text{line element}$$

2) Cylindrical coordinates.



$$\boxed{d\vec{l} = dr\,\hat{r} + r\,d\phi\,\hat{\phi} + dz\,\hat{z}} \rightarrow \text{line element.}$$

$$d\vec{A} = dA_r\,\hat{r} + dA_\phi\,\hat{\phi} + dA_z\,\hat{z}$$

$dA_r = r\,d\phi\cdot dz$ { Area 1 in figure }

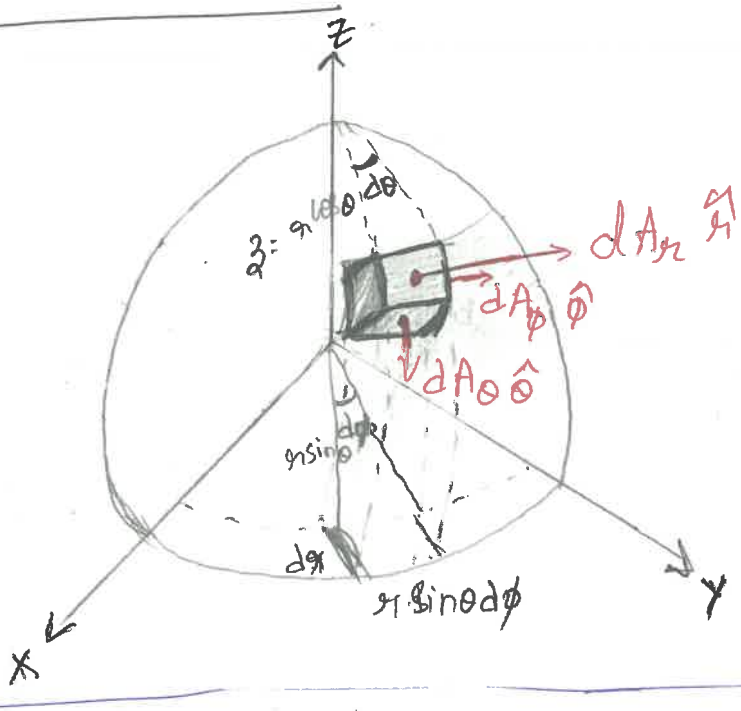
$dA_\phi = (\pm)\,dr\,dz$ { Area 2 in figure } + in +ve ϕ dirn, − in −ve ϕ dirn

$dA_z = dr\,r\,d\phi = (\pm)\,r\,dr\,d\phi$ { Area 3 in figure } + in +ve z dirn, − in −ve z dirn

$$\therefore \boxed{d\vec{A} = r\,d\phi\,dz\,\hat{r} + (\pm)\,dr\,dz\,\hat{\phi} + (\pm)\,r\,dr\,d\phi\,\hat{z}} \rightarrow \text{area element}$$

$$\boxed{dv \text{ or } d\tau = r\,dr\,d\phi\,dz} \rightarrow \text{volume element.}$$

3) Spherical Coordinates:



$$\vec{dl} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$$ → line element

$$dA_r = r\,d\theta \cdot r\sin\theta\,d\phi = r^2\sin\theta\,d\theta\,d\phi$$

$$dA_\phi = dr \cdot r\,d\theta = (\pm) r\,dr\,d\theta$$ (+ in +ve ϕ dirn, − in −ve ϕ dirn)

$$dA_\theta = dr \cdot r\sin\theta\,d\phi = (\pm) r\sin\theta\,dr\,d\phi$$ (+ in +ve θ dirn, − in −ve θ dirn)

$$\vec{dA} = r^2\sin\theta\,d\theta\,d\phi\,\hat{r} + (\pm) r\,dr\,d\theta\,\hat{\phi} + (\pm) r\sin\theta\,dr\,d\phi\,\hat{\theta}$$

↳ Area element.

$$dv = d\tau = dr \cdot r\,d\theta \cdot r\sin\theta\,d\phi$$

$$d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi$$ → volume Element

Del operator

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$ → Cartesian Coordinates

$$\nabla = \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial \phi}\hat{\phi} + \frac{\partial}{\partial z}\hat{z}$$ → Cylindrical Coordinates.

$$\nabla = \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{\phi}$$ → Spherical Coordinates

∇ is a del operator. It operates on scalars & vectors to yield different entities.

→ If we have a vector $\vec{A}$:

↳ It can be multiplied with some scalar λ to give $\lambda\vec{A}$

↳ It can be multiplied with another Vector $\vec{B}$, such that the product is a scalar:

i.e; $\vec{A} \cdot \vec{B}$

↳ It can be multiplied with another Vector $\vec{B}$, such that the product is a vector

i.e; $\vec{A} \times \vec{B}$

Similarly ∇ can operate in 3 ways:

(i) On a scalar function T: ∇ operated on T results in a vector:

$$\nabla T = \frac{\partial T}{\partial x}\hat{i} + \frac{\partial T}{\partial y}\hat{j} + \frac{\partial T}{\partial z}\hat{k}$$

Gradient of T where $T = f(x, y, z)$

(ii) On a vector $\vec{V}$: $\nabla \cdot \vec{V}$ results in a scalar

$$\therefore \nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Divergence of $\vec{V}$ where $\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$

(iii) On a vector $\vec{V}$: $\nabla \times \vec{V}$ results in a vector

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

→ Curl of a vector $\vec{V}$ where $\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$

These are written in Cartesian coordinates. Similar expressions can be written for cylindrical & Spherical coordinates using the appropriate expression for ∇.

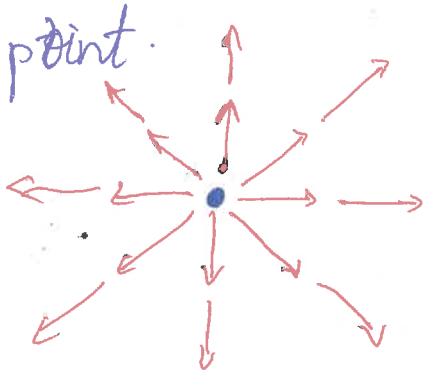
$$\nabla T = \frac{\partial T}{\partial x}\hat{i} + \frac{\partial T}{\partial y}\hat{j} + \frac{\partial T}{\partial z}\hat{k}$$ → Gradient

$$\nabla \cdot V = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$ → Divergence

$$\nabla \times V = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} = \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right)\hat{i} - \left(\frac{\partial V_z}{\partial x} - \frac{\partial V_x}{\partial z}\right)\hat{j} + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)\hat{k}$$ ↑ Curl

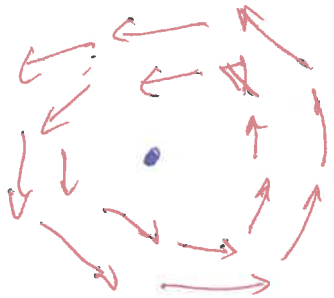
→ Divergence tells us how a vector $\vec{V}$ spreads out at a point.

eg.



→ Curl tells us how the Vector V, swirls around the point.

eg:



→ Gradient tells us the maximum increase of a function T. The magnitude gives the slope (rate of increase) along this maximum increase.

Example Problems:

1) $T = T(x, y, z)$ is the temperature in a room let $T = 20 + 2x + 3y$. find the gradient of the temperature.

Sol

$$T = 20 + 2x + 3y$$

∇T } gradient

$$\Rightarrow \nabla T = \frac{\partial T}{\partial x}\hat{i} + \frac{\partial T}{\partial y}\hat{j} + \frac{\partial T}{\partial z}\hat{k}$$

$$= \frac{\partial(20+2x+3y)}{\partial x}\hat{i} + \frac{\partial(20+2x+3y)}{\partial y}\hat{j} + \frac{\partial(20+2x+3y)}{\partial z}\hat{k}$$

$$= 2\hat{i} + 3\hat{j} + 0\hat{k}$$

$$\therefore \nabla T = 2\hat{i} + 3\hat{j}$$

$|\nabla T| = \sqrt{2^2+3^2} = \sqrt{13}$ } slope of the gradient.

2) Let the air flow inside a room be given as a vector V such that $V = x\hat{i} + y\hat{j}$. How is the wind spreading.

Sol What is effectively being asked is the divergence of the vector V.

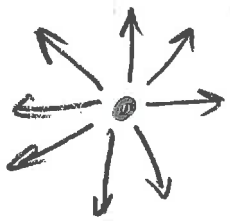
$\therefore \nabla \cdot V$ is to be determined.

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

$$= \frac{\partial (x)}{\partial x} + \frac{\partial (y)}{\partial y} + \frac{\partial (0)}{\partial z}$$

$$= 1 + 1 + 0$$

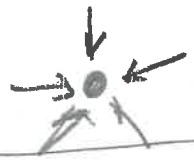
$$= \underline{\underline{2}}$$



$\because \nabla \cdot \vec{V} > 0$
It is behaving like a source. (i.e; Spreading out)

Note: If $\nabla \cdot V < 0$ } It would behave as a sink

eg:



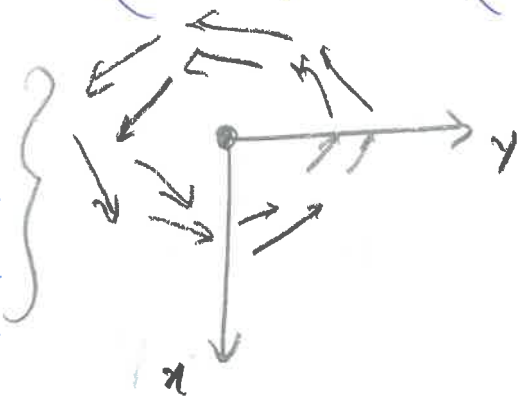
3) Determine the Curl of a vector $\vec{V} = -y\hat{i} + x\hat{j}$.

Sol

$$\nabla \times V = \text{Curl} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 0 \end{vmatrix}$$

$$= \hat{i}(0 - 0) - \hat{j}(0 + 0) + \hat{k}(1 - (-1))$$

$$= 2\hat{k}$$



4) Given that $\vec{E} = x z^3 \hat{i} - 2x^2 y z \hat{j} + 2y z^4 \hat{k}$. Find the divergence of $\vec{E}$ at (1, 2, 3).

Sol

$\nabla \cdot \vec{E}$ } divergence of E

$$= \frac{\partial (x z^3)}{\partial x} + \frac{\partial}{\partial y}(-2x^2 y z) + \frac{\partial}{\partial z}(2 y z^4)$$

$$= z^3 - 2x^2 z + 8 y z^3$$

at (1, 2, 3) $\nabla \cdot \vec{E}$ becomes

$$\nabla \cdot \vec{E} = (3^3 - 2 \cdot 1^2 \cdot 3 + 8 \cdot 2 \cdot 3^3)$$

$$= \underline{453}$$

5) Find the divergence of a position vector $\vec{r}$ in Cartesian coordinates

Sol

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\nabla \cdot \vec{r} = \text{divergence} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = \underline{1 + 1 + 1 = 3}$$

6) Find the ~~divergence~~ gradient of $\frac{1}{r}$; where $\vec{r}$ = position vector in Cartesian Coordinates.

Sol Being asked to Calculate $\nabla\left(\frac{1}{r}\right)$ where r is magnitude of $\vec{r}$ which is the position vector in Cartesian Coordinates

$$\Rightarrow \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore r = \sqrt{x^2+y^2+z^2}$$

$$\frac{1}{r} = \frac{1}{\sqrt{x^2+y^2+z^2}} \Rightarrow \nabla\left(\frac{1}{r}\right) = \nabla\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)$$

$$\nabla\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)\hat{i} + \frac{\partial}{\partial y}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)\hat{j} + \frac{\partial}{\partial z}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)\hat{k}$$

$$= \left\{\frac{-1}{2}\left(x^2+y^2+z^2\right)^{\frac{-1}{2}-1}\cdot 2x\right\}\hat{i} + \left\{\frac{-1}{2}\left(x^2+y^2+z^2\right)^{\frac{-1}{2}-1}\cdot 2y\right\}\hat{j} + \left\{\frac{-1}{2}\left(x^2+y^2+z^2\right)^{\frac{-1}{2}-1}\cdot 2z\right\}\hat{k}$$

$$= \frac{-1}{\left(\sqrt{x^2+y^2+z^2}\right)^3}\left\{x\hat{i} + y\hat{j} + z\hat{k}\right\} = \frac{-\vec{r}}{r^3}$$

$$\therefore \nabla\left(\frac{1}{r}\right) = \frac{-\vec{r}}{r^3} \text{ or } \frac{-\hat{r}}{r^2}$$

7) Find $\nabla \cdot (r^n \vec{r})$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ ∴

Sol

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow \underset{\text{magnitude}}{r} = \sqrt{x^2+y^2+z^2}$$

$$\therefore \quad r^n = (x^2+y^2+z^2)^{\frac{n}{2}}$$

$\vec{r}\, r^n$ or $r^n \vec{r}$

$$= x(x^2+y^2+z^2)^{\frac{n}{2}}\hat{i} + y(x^2+y^2+z^2)^{\frac{n}{2}}\hat{j} + z(x^2+y^2+z^2)^{\frac{n}{2}}\hat{k}$$

$\nabla \cdot (\vec{r}\, r^n)$

$$= \frac{\partial}{\partial x}\left[x(x^2+y^2+z^2)^{\frac{n}{2}}\right] + \frac{\partial}{\partial y}\left[y(x^2+y^2+z^2)^{\frac{n}{2}}\right] + \frac{\partial}{\partial z}\left[z(x^2+y^2+z^2)^{\frac{n}{2}}\right]$$

↙ using product rule of differentiation

$$\frac{\partial x}{\partial x}\cdot(x^2+y^2+z^2)^{\frac{n}{2}} + x\cdot\frac{\partial(x^2+y^2+z^2)^{\frac{n}{2}}}{\partial x}$$

$$(x^2+y^2+z^2)^{\frac{n}{2}} + x\left\{\frac{n}{2}\cdot(x^2+y^2+z^2)^{\frac{n}{2}-1}\cdot 2x\right\}$$

$$(x^2+y^2+z^2)^{\frac{n}{2}} + nx^2(x^2+y^2+z^2)^{\frac{n}{2}-1}$$

This is just the first term. Now writing this for all 3 terms we get

$\nabla \cdot (\vec{r}\, r^n)$

$$= (x^2+y^2+z^2)^{\frac{n}{2}} + nx^2(x^2+y^2+z^2)^{\frac{n}{2}-1} + (x^2+y^2+z^2)^{\frac{n}{2}} + ny^2(x^2+y^2+z^2)^{\frac{n}{2}-1}$$
$$+ (x^2+y^2+z^2)^{\frac{n}{2}} + nz^2(x^2+y^2+z^2)^{\frac{n}{2}-1}$$

$\therefore \nabla \cdot (\vec{r}\, r^n)$

$= 3 (x^2+y^2+z^2)^{\frac{n}{2}} + n (x^2+y^2+z^2)^{\frac{n}{2}-1} (x^2+y^2+z^2)$

$= 3 (x^2+y^2+z^2)^{\frac{n}{2}} + n \cdot (x^2+y^2+z^2)^{\frac{n}{2}}$

$= (3+n) (x^2+y^2+z^2)^{\frac{n}{2}} = (3+n) r^n$

$\therefore \nabla \cdot (\vec{r}\, r^n) = (3+n) r^n$